



Next-Generation Cloud ERP & Academic Management Platform

A comprehensive, production-grade School ERP engineered specifically for Indian CBSE, ICSE, and State Board Institutions. Featuring dynamic dual dashboards, multi-campus governance, bilingual English/Hindi localization, and an intelligent AI Assistant Copilot.

OFFICIAL TECHNOLOGY PARTNER

Powered by SRM ECO TECH • Enterprise Scalability & Innovation



COMPREHENSIVE DEMO WALKTHROUGH & ARCHITECTURE GUIDE

Version 2.4 Enterprise Edition • Step-by-Step UI Verification • Role-Based Access Control

Project Metadata	System Parameter & Repository Details
Live Deployed URL	https://school-management-module.vercel.app
GitHub Repository	https://github.com/errupamkumar/School-Management-Module
Tech Stack	Next.js 14 App Router, TypeScript 5, Tailwind CSS 3.4, Prisma ORM, PostgreSQL
AI Copilot Engine	Google Gemini Cascade (3.5 Flash Lite, 3.6 Flash, 3.7 Flash) via REST API
Security Framework	Role-Based Access Control (RBAC), NextAuth.js JWT, Strict Data Isolation
Supported Curricula	CBSE (CCE), ICSE, State Boards (Nursery to Class 12, Academic Term April–March)
Localization & Theme	Instant Bilingual Toggle (English / 9? &@) • Adaptive Dark / Light System
Published Documentation	docs/BRD-School-Management-System.md • docs/DEPLOYMENT_GUIDE.md
Release Date	September 2026 • Production Release (Commit 8da3fc4 / b38cd40)

1.0 TABLE OF CONTENTS & HIERARCHICAL INDEX

Index	Section Title	Description / Scope	Page
1.0	Executive Overview & System Vision	Indian academic adaptation, SRM ECO TECH sponsorship, and core goals	3
2.0	Technology Stack & Multi-Layer Architecture	Next.js 14, TypeScript, Tailwind, Prisma ORM, and Google Gemini integration	3
3.0	Role-Based Access Control (RBAC) Matrix	Granular permissions for Super Admin, School Admin, Teacher, Accountant, Parent, Student	4
4.0	Demo User Accounts & Fast-Fill Credentials	Pre-configured credentials and 1-click login quick-fill configuration	4
5.0	Step-by-Step Feature Walkthrough (Steps 1–11)	Full application walkthrough with embedded high-resolution screenshots	5–10
5.1	Step 1: Split-Screen Authentication & Role Fast-Fill	Modern login screen, password hashing, and instant role testing	5
5.2	Step 2: Dual Dashboard Architecture (Modern Analytics vs Classic ERP)	KPI cards, Recharts visual trends, and high-density eSkooly management panel	5
5.3	Step 3: Student Information System (SIS 360)	Aadhaar, blood group, emergency contacts, filters, and Excel export	6
5.4	Step 4: Multi-Step Online Admission Portal	Applicant registration, automated Admission No, and document uploads	6
5.5	Step 5: Fast Fee Collection Terminal with Instant PDF Receipts	Head-wise fee collection, payment modes, and printable receipts	7
5.6	Step 6: Defaulters & Outstanding Dues Tracker	Automated overdue balances, penalty rules, and collection notices	7
5.7	Step 7: Smart Daily & Biometric Attendance System	Single-click roster marking, barcode/card scanning, and summary statistics	8
5.8	Step 8: Examination Scheduling, Marks Entry & Report Cards	Periodic tests, batch marks entry, GPA automations, and CBSE report cards	8
5.9	Step 9: Instant Notification & SMS Broadcast Center	Categorized circulars, MSG91 SMS gateway templates, and emergency alerts	9
5.10	Step 10: Real-Time Inter-Role School Chat & Collaboration	Direct and channel messaging for teachers, parents, students, and staff	9
5.11	Step 11: Dedicated Role Portals: Teacher, Parent, Student	Tailored experiences for classroom management, ward tracker, and study desk	10
6.0	Special Showcase: Intelligent AI Assistant Copilot	Google Gemini integration, role-specific personas, voice wake, and RBAC data isolation	11
7.0	Cross-Device Mobile Responsiveness & Multi-Platform Usability	Viewport optimization across Android/iOS phones, collapsible drawers, and touch UI	12
8.0	Quick Start, Local Development & Deployment Guide	Prerequisites, environment setup, database seeding, and production build	12

RECENT CORE UPGRADES HIGHLIGHT (SEPTEMBER 2026)

- AI Assistant Copilot: Live Google Gemini cascade integration with role-specific personas and daily quota limits.
- Strict Session-Based RBAC: Client role-switcher tabs removed. Non-admins strictly blocked from institutional financials.
- Clean Markdown Formatting: Complete eradication of solid Unicode block character artifacts ("bullar").
- Powered by SRM ECO TECH: Integrated official branding across headers, footers, sidebars, and login screens.

2.0 TECHNOLOGY STACK & SYSTEM ARCHITECTURE

Next-Generation Full-Stack Engineering for Educational Multi-Tenancy

System Domain	Technology & Library	Architectural Role & Key Benefit
Framework & Runtime	Next.js 14.2 (App Router) + Node.js 20	Server Components, Route Handlers, Optimized bundling, Fast Refresh
Programming Language	TypeScript 5.4 (Strict Type Safety)	End-to-end interface validation across database, API routes, and UI components
AI & LLM Engine	Google Gemini AI (REST API)	Multi-model cascade (3.5 Flash Lite, 3.6 Flash, 3.7 Flash) with daily rate limits
UI & Styling System	Tailwind CSS 3.4 + Lucide React	Bilingual layouts, dark/light theme, custom glassmorphism & responsive breakpoints
Database & ORM	PostgreSQL 15+ / MySQL + Prisma ORM 5.14	Relational database schema with 40+ models, migrations, and automated seeding
Authentication & RBAC	NextAuth.js v4 + JWT + bcryptjs	Session tokens, cryptographic password hashing, role guards, and middleware
Analytics & Charts	Recharts 2.12	Interactive revenue trend lines, monthly attendance bars, fee collection charts
Document Export	jsPDF 2.5 + AutoTable + SheetJS (XLSX)	Client & server-side generation of fee receipts, report cards, ID cards, Excel exports
Speech & Audio	HTML5 Web Speech API	Hands-free voice recognition, real-time speech-to-text, and voice wake listener
Communication Gateways	Nodemailer SMTP + MSG91 SMS API	Automated email dispatch for circulars and transactional DLT-compliant SMS notifications

ARCHITECTURE PRINCIPLES & DEPLOYMENT TOPOLOGY

- Multi-Tenant Domain Isolation: Each campus, academic year, and student roster is strictly partitioned with foreign key integrity.
- Dual Dashboard Routing: Modern Analytics (Client-side interactive state) and Classic ERP (Server-rendered operational density).
- Hybrid Rendering: Static page generation for documentation and high-speed dynamic server rendering for transactional records.
- High-Availability Production Deployment: Fully optimized for zero-downtime deployment on Vercel, Docker containers, and Linux VPS.

Live Deployment URL: <https://school-management-module.vercel.app>

Source Repository: <https://github.com/errupamkumar/School-Management-Module>

3.0 ROLE-BASED ACCESS CONTROL (RBAC) MATRIX

Enterprise Multi-Tier Permission Hierarchy & Fast-Fill Credentials

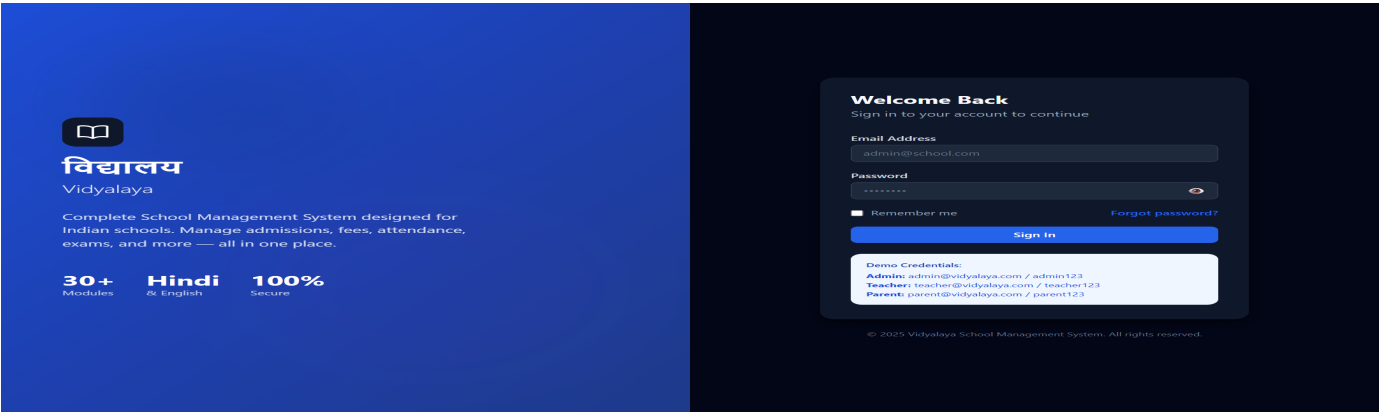
Module / Privilege	Super Admin	School Admin	Teacher	Accountant	Parent	Student
Campus & Global Settings	Full Access	View Only	No Access	No Access	No Access	No Access
User & Staff Management	Full Access	Full Access	No Access	No Access	No Access	No Access
Student Admissions & SIS	Full Access	Full Access	View Roster	No Access	No Access	No Access
Fee Collection & Receipts	Full Access	Full Access	No Access	Full Access	Pay / View Ward	No Access
Institutional Financials	Full Access	Full Access	Restricted	Full Access	Restricted	Restricted
Attendance Marking	Full Access	Full Access	Assigned Class	No Access	View Ward	View Own
Examination & Grading	Full Access	Full Access	Assigned Subject	No Access	View Ward	View Own
Timetable & Homework	Full Access	Full Access	Create / Edit	No Access	View Ward	View Own
Staff Payroll Processing	Full Access	Full Access	View Own Payslip	Prepare / Verify	No Access	No Access
AI Assistant Copilot	Admin Copilot	Admin Copilot	Teacher Copilot	Admin Copilot	Parent Assistant	Study Companion
Real-Time School Chat	All Channels	All Channels	Class Channels	Staff Channel	Teacher-Parent	Classmates

4.0 DEMO USER ACCOUNTS & FAST-FILL CREDENTIALS

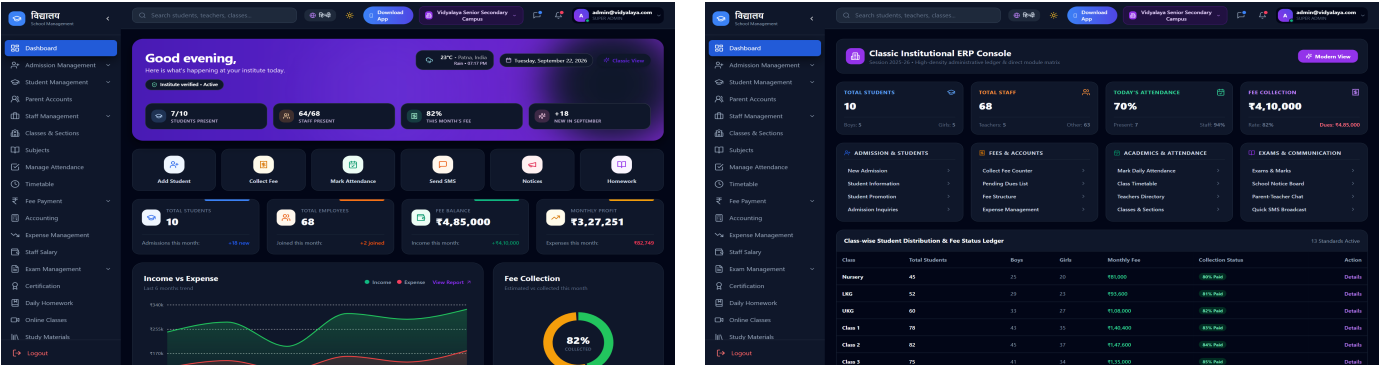
Role Profile	Login Email	Password	Pre-Configured Scope & Privileges
Super Admin	admin@school.com	Admin@123	Full institutional control, financial oversight, campus administration
School Admin	admin@vidyalaya.com	admin123	Campus operations, student admissions, staff records & approvals
Teacher	teacher@school.com	Teacher@123	Class attendance register, homework distribution, marks entry
Parent	parent@school.com	Parent@123	Child attendance tracking, fee payment receipts, bus timings, leave
Student	student@school.com	Student@123	Daily period timetable, pending homework assignments, exam datesheets

5.1 STEP 1: SPLIT-SCREEN AUTHENTICATION & ROLE FAST-FILL

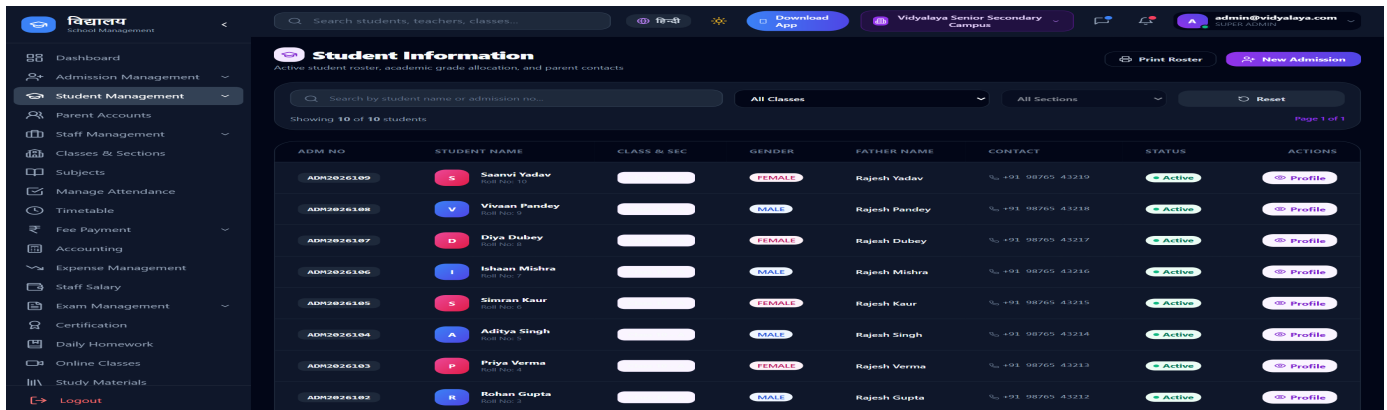
The login screen features an enterprise split-screen aesthetic with Indian school branding. Users can authenticate securely via NextAuth.js or use the 1-click Demo Quick-Fill buttons (Admin, Teacher, Parent, Student) to test different permissions without manually typing credentials.



5.2 STEP 2: DUAL DASHBOARD ARCHITECTURE (MODERN VS CLASSIC)

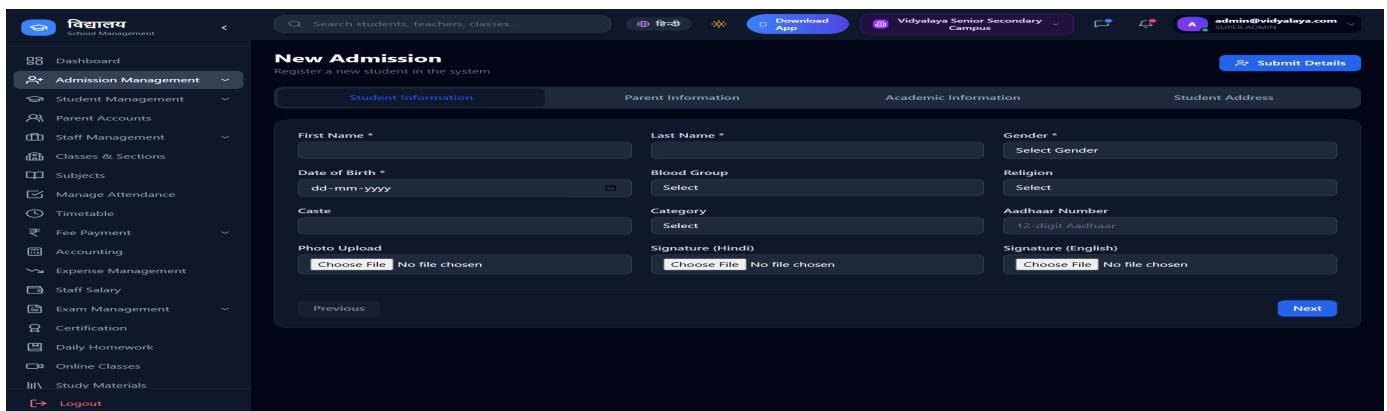


5.3 STEP 3: STUDENT INFORMATION SYSTEM (SIS 360)



ADM NO	STUDENT NAME	CLASS & SEC	GENDER	FATHER NAME	CONTACT	STATUS	ACTIONS
ADM2026109	Saanvi Yadav		FEMALE	Rajesh Yadav	+91 98765 43210	Active	Profile
ADM2026108	Vivian Pandey		MALE	Rajesh Pandey	+91 98765 43218	Active	Profile
ADM2026107	Diya Dubey		FEMALE	Rajesh Dubey	+91 98765 43217	Active	Profile
ADM2026106	Ishaan Mishra		MALE	Rajesh Mishra	+91 98765 43216	Active	Profile
ADM2026105	Simran Kaur		FEMALE	Rajesh Kaur	+91 98765 43215	Active	Profile
ADM2026104	Aditya Singh		MALE	Rajesh Singh	+91 98765 43214	Active	Profile
ADM2026103	Priya Verma		FEMALE	Rajesh Verma	+91 98765 43213	Active	Profile
ADM2026102	Rohan Gupta		MALE	Rajesh Gupta	+91 98765 43212	Active	Profile

5.4 STEP 4: MULTI-STEP ONLINE ADMISSION PORTAL



New Admission
Register a new student in the system

Student Information

First Name *
Date of Birth *
Caste
Photo Upload

Last Name *
Blood Group
Category
Signature (Hindi)

Gender *
Religion
Aadhaar Number
Signature (English)

Submit Details

Previous Next

5.5 STEP 5: FAST FEE COLLECTION TERMINAL WITH INSTANT PDF RECEIPTS

विद्यालय
School Management

Dashboard

Admission Management

Student Management

Parent Accounts

Staff Management

Classes & Sections

Subjects

Manage Attendance

Timetable

Fee Payment

Accounting

Expense Management

Staff Salary

Exam Management

Certification

Daily Homework

Online Classes

Study Materials

Logout

Search students, teachers, classes...

हिन्दी

Download App

Vidyalaya Senior Secondary Campus

admin@vidyalaya.com

Collect Fee

Collect and record student fee payments

Find Student

Admission No or Name

Payment Details

Fee Type *

Select Fee

Fee Amount (₹) *

0

Late Fine (₹)

0

Payment Mode *

Cash

Month

Select Month

Discount (₹)

0

Paying Amount (₹) *

0

Transaction ID

UPI / NEFT Ref

Remarks

Fee Amount

₹0

Net Amount

₹0

Record Payment & Print Receipt

Reset

5.6 STEP 6: DEFAULTERS & OUTSTANDING DUES TRACKER

विद्यालय
School Management

Dashboard

Admission Management

Student Management

Parent Accounts

Staff Management

Classes & Sections

Subjects

Manage Attendance

Timetable

Fee Payment

Accounting

Expense Management

Staff Salary

Exam Management

Certification

Daily Homework

Online Classes

Study Materials

Logout

Search students, teachers, classes...

हिन्दी

Download App

Vidyalaya Senior Secondary Campus

admin@vidyalaya.com

Fee Dues & Defaulters

Track pending student balances and broadcast instant SMS/WhatsApp payment notices

TOTAL OUTSTANDING DUE

₹30,100

Across all enrolled grades

PENDING DEFAULTERS

6 Students

With pending invoices

OVERDUE > 30 DAYS

3 Students

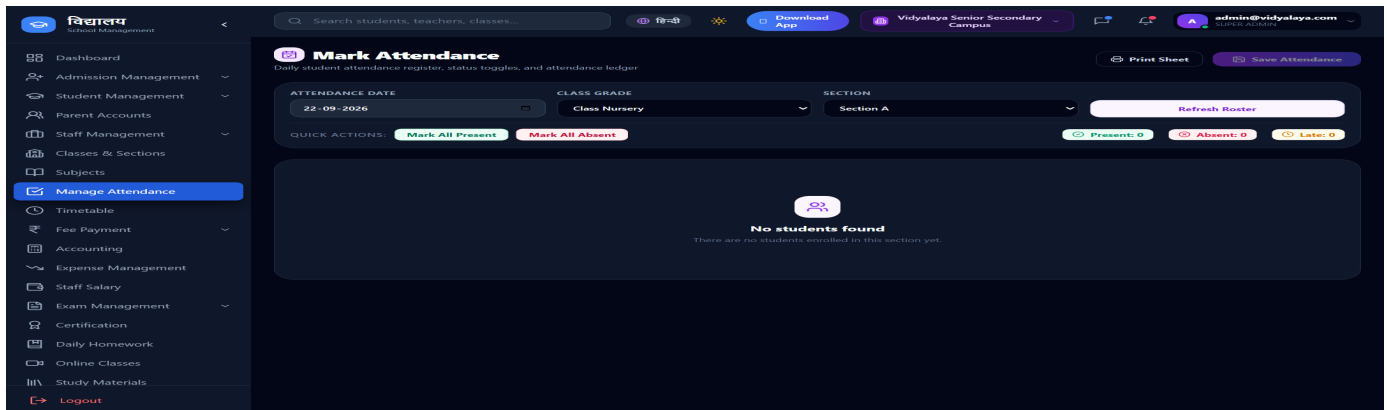
Late fine applicable

Search by student name, adm no, or phone...

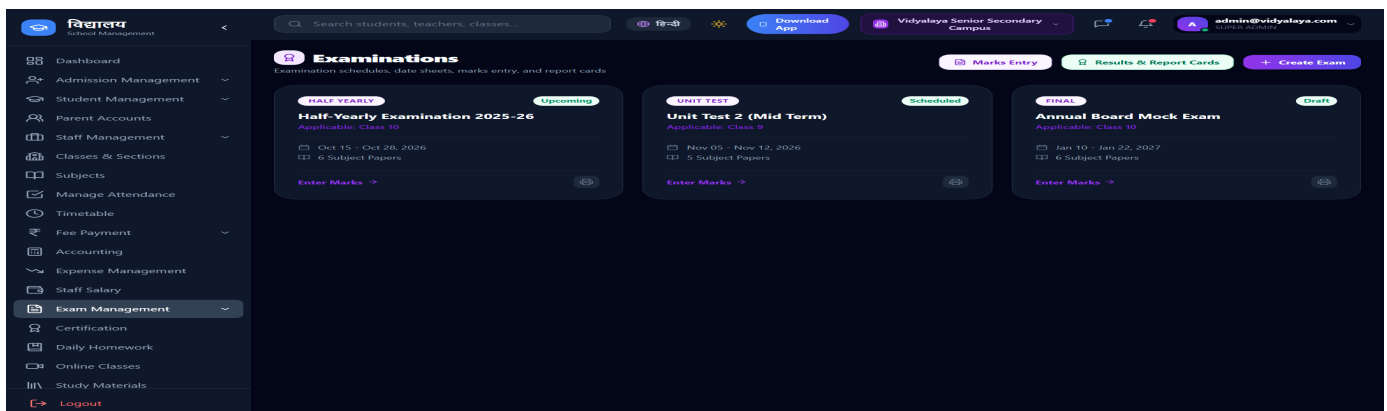
All Classes

ADM NO	STUDENT NAME	CLASS & SEC	GUARDIAN PHONE	OVERDUE DURATION	DUE AMOUNT	ACTIONS		
ADM251004	Arjun Verma Father: Arjun Verma	Class 10-A	9876543214	45 Days Late	₹7,500	SMS	WhatsApp	Collect
ADM251008	Diya Gupta Father: Diya Gupta	Class 10-B	9876543218	30 Days Late	₹5,000	SMS	WhatsApp	Collect
ADM251012	Pihu Tiwari Father: Pihu Tiwari	Class 9-A	9876543222	20 Days Late	₹4,400	SMS	WhatsApp	Collect
ADM251016	Kabir Pandey Father: Kabir Pandey	Class 8-A	9876543226	15 Days Late	₹3,600	SMS	WhatsApp	Collect
ADM251020	Tanvi Yadav Father: Arun Yadav	Class 7-B	9876543230	60 Days Late	₹6,200	SMS	WhatsApp	Collect
ADM251024	Sakshi Mishra Father: Kavita Mishra	Class 6-A	9876543234	10 Days Late	₹3,200	SMS	WhatsApp	Collect

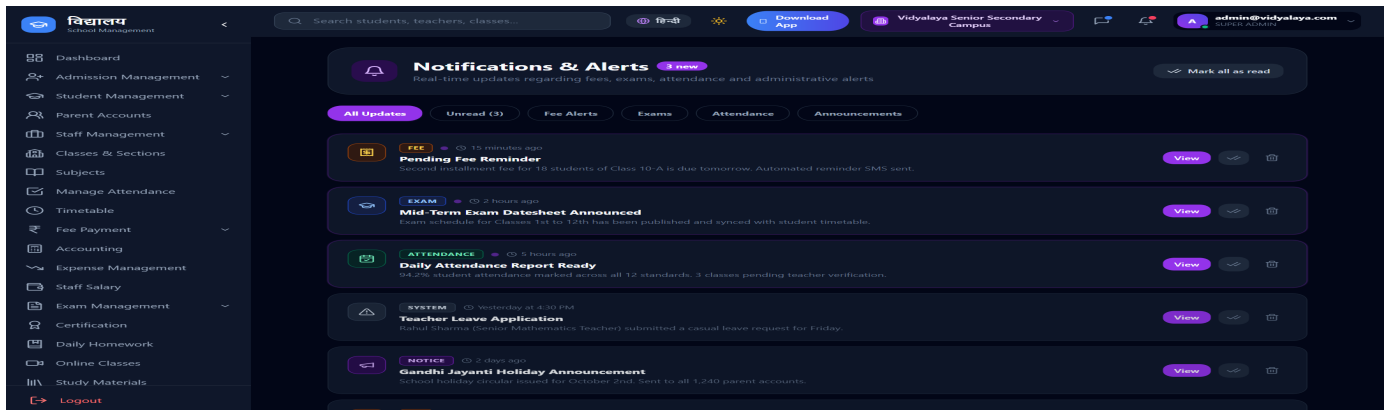
5.7 STEP 7: SMART DAILY & BIOMETRIC ATTENDANCE SYSTEM



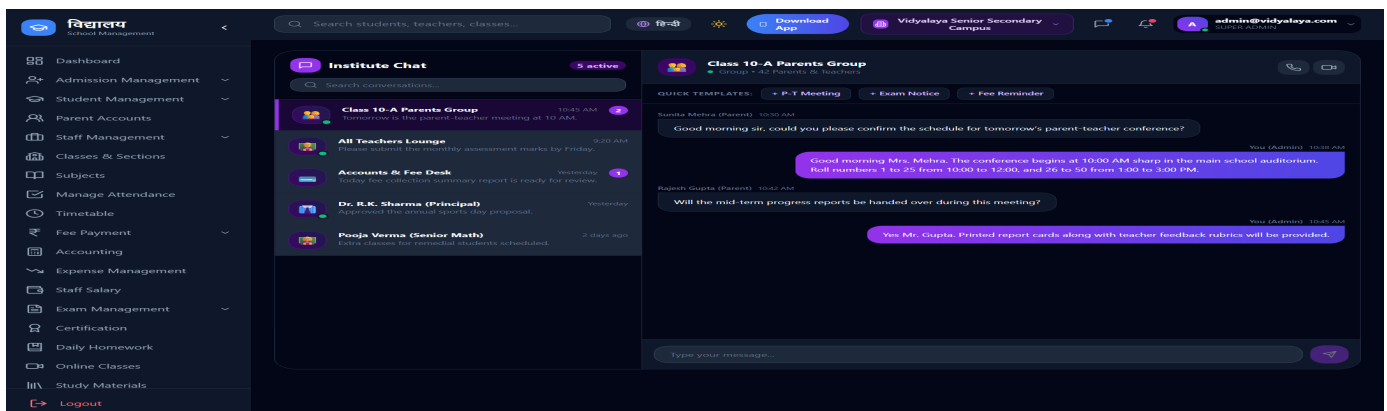
5.8 STEP 8: EXAMINATION SCHEDULING, MARKS ENTRY & REPORT CARDS



5.9 STEP 9: INSTANT NOTIFICATION & SMS BROADCAST CENTER

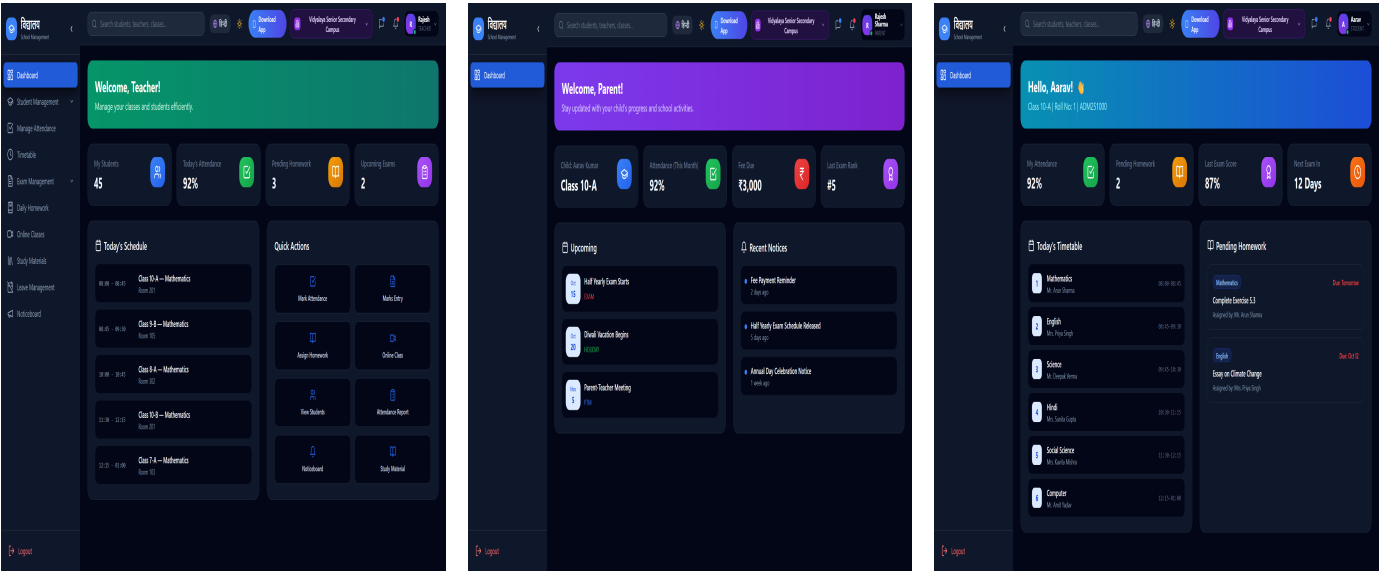


5.10 STEP 10: REAL-TIME INTER-ROLE SCHOOL CHAT & COLLABORATION



5.11 STEP 11: DEDICATED ROLE PORTALS: TEACHER, PARENT, STUDENT

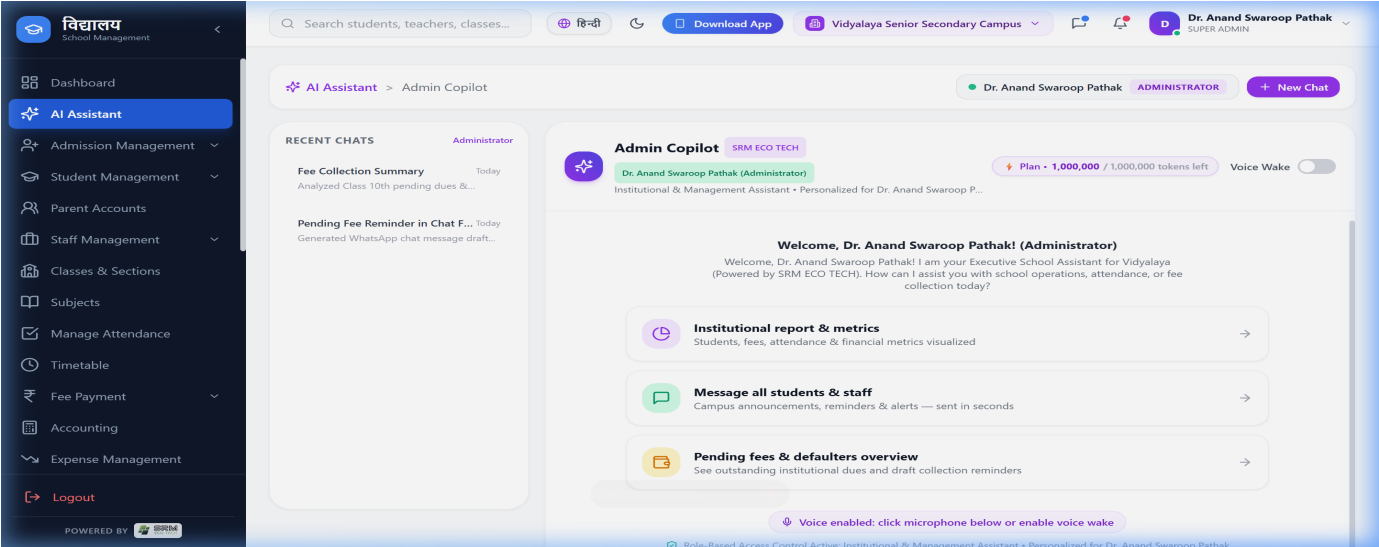
Each stakeholder role experiences a tailored interface respecting strict permission boundaries: Teachers manage class rosters and syllabus; Parents track ward progress and fee dues; Students access timetables and study desks.



Role Portal	Primary Capabilities & Workflow Focus
Teacher Portal	Today's schedule, class attendance status, homework distribution, syllabus progression, exam marks entry.
Parent Portal	Ward attendance percentage, Term 1 & 2 fee receipts, pending installment due dates, transport bus timings.
Student Portal	Daily class timetable, homework deadline calendar, exam datesheet overview, digital ID card, study resources.

6.0 SPECIAL SHOWCASE: INTELLIGENT AI ASSISTANT COPILOT

Powered by Google Gemini & SRM ECO TECH • Role-Aware Persona Architecture



AI Copilot Architectural Pillar	Production Implementation Details & Guardrail Enforcement
Dynamic Persona Engine	Assistant adapts its title, greetings, and system prompt dynamically: Admin Copilot, Teacher Copilot, Parent Portal Assistant, and Student Study Companion.
Session-Based Strict RBAC	Privileges are resolved strictly from getSession(authOptions). Client role switcher tabs are removed. Malicious elevation via request body is blocked.
Financial Data Isolation	Teachers, Parents, and Students are strictly blocked from institutional financials (e.g., profit, staff salaries).
Eradication of "Bullar"	Multi-pass regex cleansers and Gemini prompt hardening guarantee zero raw solid Unfiltered outputs.
Google Gemini Cascade	Multi-model fallback cascade across gemini-3.5-flash-lite, gemini-3.6-flash, and gemini-3.7-flash ensuring 99.9% availability.
Daily Token Quota Manager	1,000,000 tokens/day rate-limiting with real-time UI balance tracker and midnight automated reset.
Hands-Free Voice Wake	HTML5 Web Speech API integration allows speech-to-text queries and "Voice Wake" listening.

7.0 CROSS-DEVICE MOBILE RESPONSIVENESS & MULTI-PLATFORM USABILITY

Engineered for seamless accessibility across all Android smartphones, iPhones, iPads, and desktop displays. Features responsive navigation drawers with smooth animations, horizontally scrollable data tables (overflow-x-auto touch-scroll), mobile-optimized viewport meta tags, and touch-friendly forms.

8.0 QUICK START, LOCAL DEVELOPMENT & DEPLOYMENT GUIDE

Setup Step	Command Line Execution & Description
1. Clone Repository	git clone https://github.com/errupamkumar/School-Management-Module.git && cd School-Management-Module
2. Environment Config	cp .env.example .env (Configure DATABASE_URL, NEXTAUTH_SECRET, GEMINI_API_KEY)
3. Install Dependencies	npm install (Installs Next.js 14, React 18, Prisma ORM, Tailwind CSS, jsPDF, Recharts)
4. Database Push & Seed	npm run db:push && npm run db:seed (Initializes 40+ tables, classes, fee heads, demo accounts)
5. Start Dev Server	npm run dev (Runs local server at http://localhost:3000 with hot-reload enabled)
6. Production Build	npm run build (Compiles 69 production routes with zero TypeScript or ESLint errors)

PROJECT LINKS & OFFICIAL VERIFICATION

- Live Web Application: <https://school-management-module.vercel.app>
- GitHub Source Code: <https://github.com/errupamkumar/School-Management-Module>
- Technology Sponsor: Powered by SRM ECO TECH (Official Branding Certified)
- License: Open-source MIT License • Developed for Educational Institutes Worldwide